Let's Measure Agreement, Consensus, and Polarization in Ordinal Data

Clem Aeppli (Harvard University, ORCID: 0000-0002-4003-7938)
Didier Ruedin (University of Neuchâtel, and University of the Witwatersrand, ORCID: 0000-0001-5213-4316)

Corresponding author:
Didier Ruedin
University of Neuchâtel
Rue Abram-Louis-Breguet 2
2000 Neuchâtel, Switzerland
didier.ruedin@unine.ch

Abstract

There are different measures to capture agreement, consensus, concentration, dispersion, and polarization in ordinal data. To better understand how different measures work in practice, we compare consensus scores across specific situations: constructed cases, simulated data where we know the underlying distribution, and empirical data. Researchers have addressed the 'problem' of measuring agreement, consensus, and polarization in ordinal data in many ways across many subfields; we highlight the similarities as well as the distinctive features of these approaches. In many cases, the different measures are highly correlated, but the choice of method can lead to substantially different conclusions in situations that are difficult for non-mathematical users to predict. Unless researchers prioritize specific properties, we recommend a combination of measures and graphics to qualitatively examine the distribution.

Keywords: agreement, consensus, polarization, ordinal data, measurement

Authors' statement: Authors are listed alphabetically. DR designed the study, CA and DR carried out the simulations, CA and DR carried out the analysis, DR and CA wrote the paper.

Funding: This work was supported by the Brown University LINK Internship Award [to CA]; the NCCR on the move [grant numbers 182897 and 205605]; and the Swiss National Science Foundation [grant number 200939].

Data availability statement: Replication material will be available on OSF. Data on the politicization of migration are available at https://doi.org/10.7910/DVN/IEKX8U.

Conflicts of interest: The authors declare that there are no conflicts of interest to disclose.

1. Introduction

In the social sciences, we often encounter ordinal categorical data – just think of Likert survey items ranging from 'very strongly agree' to 'very strongly disagree,' or ordinal categorical data produced in content analysis when coding the position of a text. Like others before us, we wanted to know to what extent positions are concentrated or dispersed. For instance, Krymkowski et al. (2009) sought to capture the degree of 'norm crystallization' among outdoor recreators, or Blair and Lacy (2000) compared the degree of consensus over belief in God between Black and White Americans. Harding (2011) and Berg et al. (2013) examined how cultural heterogeneity affects educational attainment. Indeed, disagreement and consensus of public opinion are mainstay of the social sciences (e.g., Balietti et al. 2021), and scholars regularly assess the degree of ideological polarization from Likert-type survey responses (e.g., Alwin and Tufiş 2016; Enders 2021). The need to accurately measure disagreement has become even more pressing in an era marked by political polarization (Kubin and von Sikorski 2021; Levendusky 2009; Lupu 2015).

The sample standard deviation can be used to measure the concentration and dispersion of continuous data. But when faced with ordinal data, methodologically versed researchers often hesitate to use the standard deviation. If we use the sample standard deviation for ordinal data, we assume that the categories are equally spaced, an assumption that is often violated or undetermined. Indices of qualitative variation (IQV) developed to assess dispersion in nominal data are equally unsuitable because, by design, they fail to consider the important ordered aspect of data. In a disparate literature spanning several disciplines,

researchers have developed different measures of ordinal concentration and dispersion. The range of terminology employed demonstrates that such measures are often developed isolated from one another. Various measures refer to ‘consensus’ (Leik 1966; Davies 1970; Tastle and Wierman 2007), ‘concentration’ (Blair and Lacy 2000), or ‘agreement’ (Van der Eijk 2001) to describe the clustering of responses.

As researchers, this diversity of approaches leaves us unsure how to proceed. We need to know which of the different approaches is the most appropriate, and to do so, we need to understand how they behave as well as how bad those violations of assumptions are in practice in the case of the sample standard deviation.

We present different measures and show how they react differently to data modifications. We use one pair of terms: ‘concentration’ and its complementary concept ‘dispersion.’ Concentration exists when many responses cluster around a particular point, whereas dispersion occurs when responses are spread out or scattered. Our choice highlights the fact that ordinal data need not reflect the opinions of respondents but may refer to positions more generally. To better understand how different measures behave in practice, we provide axiomatic scenarios, simulated ordinal data from latent continuous variables, and real-world examples. We show that two groups of measures exist that can lead to different results. The differences are difficult to anticipate for non-mathematicians because they depend on the precise coding of ordinal data. However, the contrived examples and simulations allow us to identify patterns that may guide researchers in choosing a particular measure. In the absence of prioritization by researchers, we recommend a combination of measures and highlight the benefit of qualitatively assessing distributions using graphics.

2. Measurement of concentration and dispersion

As we often do in the social sciences, researchers can treat an ordinal variable as continuous, and we can use the sample standard deviation or other measures such as the interquartile range to measure concentration and dispersion. When doing so, we assume that categories are equally spaced, and by using the standard deviation, researchers (implicitly) defend this assumption as reasonable. Equal spacing means that the response categories are evenly spaced in a substantive sense: that category 1 is as far from category 2 as category 2 is from category 3. Because we know that this assumption does not necessarily hold up or because typically we cannot test it, researchers often hesitate. For example, Blair and Lacy (2000) argue that the 'true' value of categories may follow a logarithmic pattern with scores of ln(1), ln(2), ln(3) instead of 1, 2, 3. Under this assumption, the categories move 'closer' together as they increase in the index, and the categorical standard deviation can give misleading or indirect answers.

Van der Eijk (2001) notes that if the mean of an ordinal distribution is close to either of the endpoints of the scale, the standard deviation is greatly influenced by additional points at the other extreme of the scale, even if there are few such points. Moreover, the standard deviation is difficult to use as an index because it has no natural bounds. While other measures vary between the meaningful bounds 0 and 1, with 0 corresponding to maximum dispersion and 1 to maximum concentration, the standard deviation does not.

Trying to avoid these assumptions, which are often untestable, we can use a measure of nominal dispersion, notably one of the indices of qualitative variation (IQV, Berry and Mielke 1992). Because IQV are designed to measure dispersion in unordered, nominal

data, they solve the problem of equal spacing, but we lose the important information of order, which is often central in social science research. A five-category distribution of (0, 30, 40, 30, 0), with a clear concentration at the center, receives the same IQV score as the dispersed distribution (30, 30, 0, 0, 40), which is undesirable with ordinal data.

Therefore, alternative measures have been introduced. When developing alternative measures of ordinal concentration, some scholars first identify the point of maximal concentration as any distribution in which all responses fall into a single category (Leik 1966; Van der Eijk 2001). Maximal dispersion is defined as the distribution in which half of the responses are in each extreme category, a distribution Van der Brug et al. (2015) refer to as polarization. Leik (1966), Kvålseth (1988), Blair and Lacy (2000), and others have also noted that a measure of ordinal concentration should equal 1 at the point of maximal concentration and 0 at the point of maximal dispersion. This standardization allows comparisons, and a measure of ordinal dispersion can be defined as 1 less the measure of concentration, ranging from 0 at the point of maximal concentration to 1 at the point of maximal dispersion. Considering this possibility of converting between the two, in the following we focus on concentration.

Measures take the form of algorithms, as in Van der Eijk's (2001) measure of agreement *A*, or of formulas. There are two common methods by which researchers develop formulas to measure ordinal concentration, attempting to maintain the assumptions of ordinal data while avoiding additional assumptions of distance or category width. Tastle and Wierman (2007) draw on Shannon entropy to define a measure of consensus *Cns* that involves calculating the distance between each observation and the mean. A second category of responses uses the empirical cumulative mass function (Leik 1966; Blair and Lacy 2000;

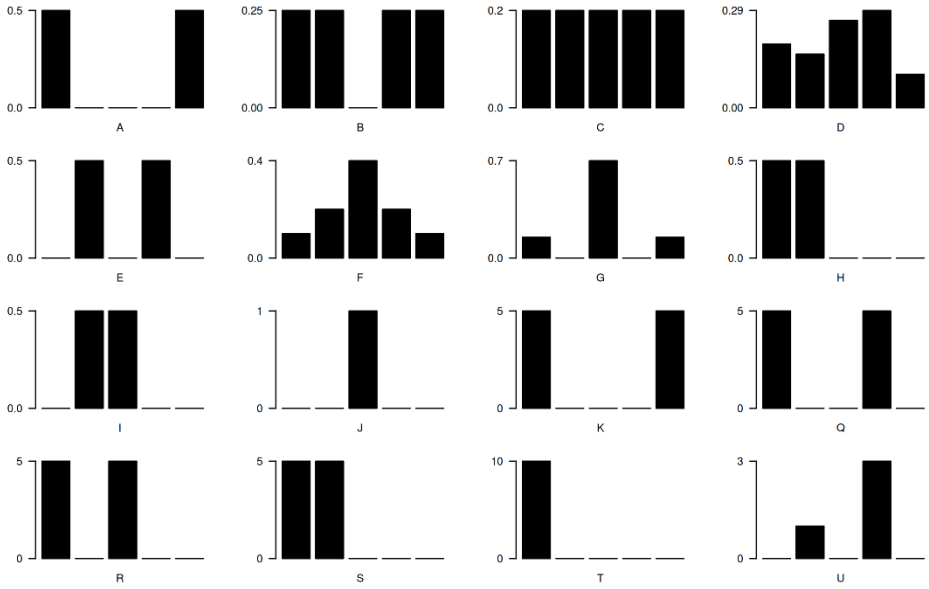
Kvålseth 1995). Davies (1970) has remarked that a cumulative-mass approach, as far as it is used in Leik's (1966) measure, risks reintroducing assumptions of equal distance between categories. We cover the technicalities of the different measures in **Appendix 1** and show that many of them are specific instances of a generalized family of measures that we call the L_N family. **Appendix 2** provides more detail on Van der Eijk's (2001) algorithm.

3. Simulated Data

Contrived examples by Tastle and Wierman

To evaluate their measure of consensus *Cns*, Tastle and Wierman (2007, 7) provided several sets of ideal typical distributions that do not assume a latent continuous variable. These cases help us to understand the dynamics of the different measures where there is progression between cases, but they are also useful for understanding cases close to the extremes. We use these to gage how different measures react to different distributions – not in the sense of a contest, but in the sense of understanding the different measures in practice. To make the distributions more accessible, we show them graphically in **Figure 1**. The corresponding concentration scores are presented in **Table 1**. These contrived examples address the concern that the continuous standard deviation does not always accurately capture what researchers mean by concentration. For example, in a highly skewed distribution where most of the mass is near a single extreme, such as distribution T in **Figure 1**, individual observations at the opposite extreme may have an outsize effect on the standard deviation because of their great distance from the mean.

Figure 1. Ideal-typical distributions from Tastle and Wierman



*Notes: Graphical presentation of the ideal-typical distributions used by Tastle & Wierman, which we labeled consecutively as A to Y. Cases A – J are from Table 1 in Tastle & Wierman. Their Table 2 includes scaled versions of case K included in this figure (cases K to N in **Appendix 3**, distribution O in the appendix differs from K by only 1%); their Table 3 corresponds to the series {K, Q, R, S, T} (distribution P is equal to distribution K). Cases U to Y from Table 4 in Tastle & Wierman are scaled versions of the distribution U included in this figure (compare **Appendix 3**). Case D does not represent an ideal-typical distribution but is one of their examples.*

Case A in Figure 1 shows ideal-typical dispersion with all observations in the extreme categories furthest left or right. Values of concentration should increase as this dispersion is reduced in the progression toward cases B and C, where the mean position is held constant. We can see in Table 1 that this is the case for all measures considered, and we note that the values of the standard deviations decrease with increasing concentration: they all have face validity. Similarly, all measures identify case J as having the maximum concentration. However, the ostensible midpoint between concentration and dispersion varies depending on the measure, and only measure *A* has an interpretable midpoint at 0.5. This means that studies interested in classifying outcomes in absolute

terms – a tendency toward concentration versus a tendency toward dispersion – find it difficult to do so.

Table 1. Concentration scores for the distributions shown in Figure 1.

	A	L1	L2	L2^2	Cns	SD	SD (log)
A 	0	0	0	0	0	2.010	0.809
B 	0.431	0.250	0.354	0.125	0.292	1.589	0.634
C 	0.500	0.400	0.447	0.200	0.434	1.421	0.571
D 	0.603	0.485	0.539	0.291	0.518	1.276	0.539
E 	0.583	0.500	0.707	0.500	0.585	1.005	0.348
F 	0.675	0.600	0.632	0.400	0.634	1.101	0.437
G 	0.712	0.700	0.700	0.490	0.700	1.101	0.463
H 	0.875	0.750	0.866	0.750	0.807	0.503	0.348
I 	0.875	0.750	0.866	0.750	0.807	0.503	0.204
J 	1.000	1.000	1.000	1.000	1.000	0.000	0.000
K 	0.000	0.000	0.000	0.000	0.000	2.108	0.848
Q 	0.292	0.250	0.500	0.250	0.322	1.581	0.731
R 	0.583	0.500	0.707	0.500	0.585	1.054	0.579
S 	0.875	0.750	0.866	0.750	0.807	0.527	0.365
T 	1.000	1.000	1.000	1.000	1.000	0.000	0.000
U	0.792	0.750	0.791	0.625	0.686	1.000	0.347

*Notes: see **Figure 1** for a description of the distributions; standard deviations are calculated under the assumption that the distributions refer to values 1, 2, 3, 4, 5, respectively. For distributions A to J, we assume 100 observations, as in Tastle & Wierman. Case C reflects a situation defined as 0.5 by Van der Eijk, while the other measures do not have such a clearly identified midpoint. Not shown are the scaled versions of distributions K and U for which only the standard deviation-based measures vary (**Appendix 3**).*

According to most measures, case F with a clear peak is more concentrated than case E. However, in this case, the standard deviation suggests no difference in concentration between the case with a single peak and that with two peaks. This is an ideal-typical instance

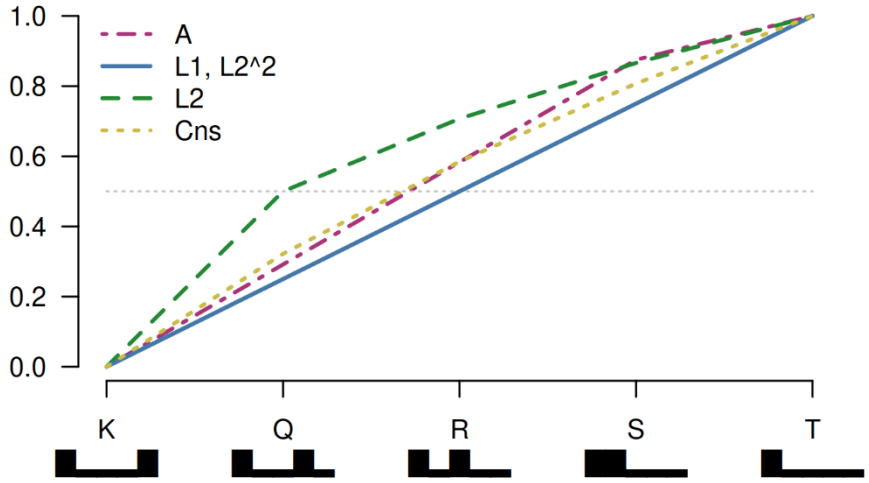
that we have not seen discussed previously, where the standard deviation fails face validity as a measure of concentration. Similarly, most measures indicate greater concentration in case G than in case F because the share in the middle category is larger; the standard deviation does not reflect this.

Cases H and I can be considered as equal concentrations, and all the measures considered yield the same result, including the standard deviation, further highlighting face validity and high correlations when considering ideal-typical cases. However, if we remove the assumption of equal spacing by taking the logarithm of the values, the results for the standard deviation vary. Similarly, where all observations fall into the same category as in case J, all measures indicate maximum concentration (and zero standard deviation). Cases K to N in **Appendix 3** refer to case K – half the observations in both extremes – but with an increasing number of observations; cases U to Y in **Appendix 3** represent another series where the number of observations increases, but the distribution across categories remains the same. Only the standard deviation reacts to the change in the number of observations, which makes comparisons in concentration difficult across samples. The other measures do not share this undesirable property.

Case O in **Appendix 3** changes 1% of the observations from one extreme category to the other. For 3 of the measures (A, L1, L2) and the standard deviation, this change is reflected in a change of number at the three-digit level, highlighting that the different measures vary in the extent to which they react to minor differences in the data. These different reactions come to the fore in the series of cases {K, Q, R, S, T} in **Figure 2**. The series illustrates a change in distributions from dispersed in case K to concentrated in case T. All the measures reflect this change, providing face validity, but the size of the steps

between cases. L2 reacts most quickly to changes away from maximum dispersion (case K on the left of the figure). Cns and A react faster than the others (L1, L2^2), where changes seem linear. We note that changes away from the maximum concentration (case T on the right of the figure) are not symmetrical.

Figure 2. Changes in concentration in simulated response distributions



*Notes: Concentration is given on the y-axis. See **Figure 1** for a description of each case.*

The distributions in **Table 1** are all based on 5 categories. This is an important detail to keep in mind. It follows from the definitions of each measure that splitting one of the categories into several categories generally changes the measured concentration. To illustrate this point, if we take distribution B and divide the empty middle category into three empty categories (case B7 in **Appendix 4**), all measures give numbers that differ from case B. Given that we are looking at ordinal data, it can easily be argued that dividing a category into subcategories should make no difference – after all, case B7 may just (arbitrarily) make finer distinctions. Similarly, we note that dividing the flat distribution in case C into 7 categories affects all measures except for measure *A*, which is designed to

have a fixed and interpretable midpoint (**Appendix 4**). To reiterate, scores cannot be compared when the number of categories varies, which includes recoding ordinal data into a smaller number of categories to simplify data or address small counts in some categories.

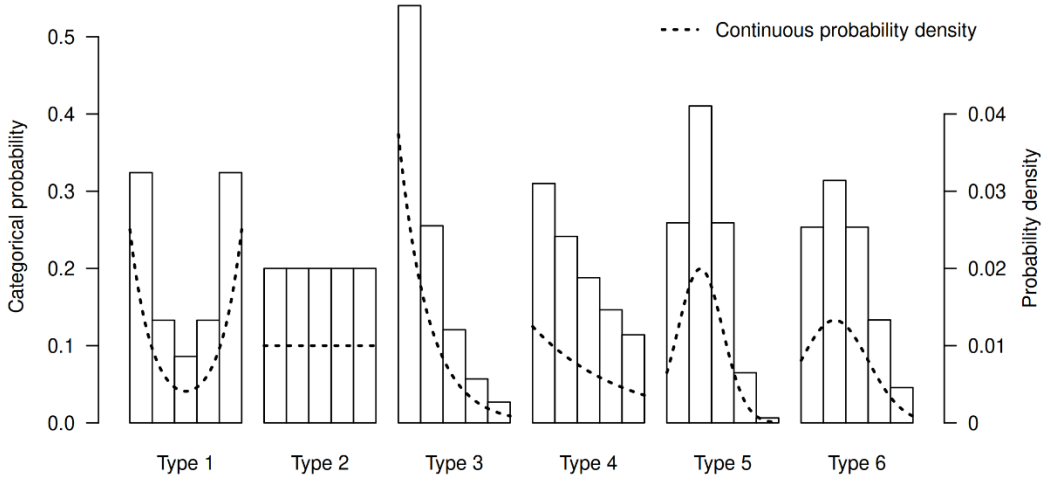
Correlations in these ideal-typical cases are remarkably high: $r \geq 0.93$ [0.85, 0.97] for all combinations that do not include the standard deviation divided by the number of observations (SD/n) or the logarithm (**Appendix 5**). The absolute value of the correlations with the standard deviation is high for all measures – 0.96 [0.92, 0.98] for *A*, 0.97 [0.93, 0.99] for L1, 0.96 [0.92, 0.98] for L2, 0.99 [0.97, 1.00] for L2^2, and 0.98 [0.96, 0.99] for *Cns* – even though some of them were designed to illustrate the weaknesses of the standard deviation for measuring concentration and dispersion in ordinal data.

Simulations assuming continuous latent variables

In the second set of simulations, we assume a continuous latent variable that is divided into categories. This provides us with a reference point that was not available in the preceding section. We simulate responses by drawing 1,000 samples of size 500 from each of the four continuous distributions on the interval [0, 100]. We simulate 6 distinct types of distribution, as illustrated in **Figure 3**. We have chosen these types of distributions because they represent distributions often found in social science literature, particularly in situations where measures of distribution or concentration are of interest. Type 1 is a bimodal distribution defined as the sum of an exponential distribution with rate parameter $\lambda=0.05$ and the reflection of this distribution about the point $x=50$. Type 2 is the uniform distribution defined on [0, 100]. Type 3 is an exponential distribution with rate parameter $\lambda=0.0375$, representing a unimodal distribution with small variance, whereas type 4 is an exponential distribution with rate parameter $\lambda=0.0125$, representing a unimodal

distribution with larger variance. Type 5 is a unimodal distribution with small variance, defined by a normal distribution with a mean of 30 and standard deviation 20, while Type 6 is a unimodal distribution with larger variance, defined by a normal distribution with a mean of 30 and standard deviation 30.

Figure 3. Simulated response distributions



Notes: The dashed line gives the continuous probability density for 6 different simulated response distributions (Type 1, Type 2, ..., Type 6); the bars break the probability density into a five-level ordinal variable.

Once the continuous distribution is simulated, we break the data into five-level ordinal variables with equal spacing. This corresponds, for instance, to the commonly used Likert scales, written {1, 2, 3, 4, 5}: a discretized version of the underlying continuous variable as it is often used in the social sciences to capture agreement at 5 distinct positions from "strongly agree" to "strongly disagree". All points in the interval [0, 20] correspond to response 1, those in interval [20, 40] correspond to response 2, and so on. In this sense, category frequencies are qualitative reflections of the continuous latent variable. After breaking the sample into these five bins, we calculate Leik's *D*, Blair and Lacy's *l*, Tastle and Wierman's *Cns*, Van der Eijk's *A*, and the categorical standard deviation. In these

simulations, as in the others that follow, we linearly transform A so that it ranges between 0 and 1, with 1 corresponding to the maximum concentration. The choice of five bins allows a direct comparison with the examples in Van der Eijk (2001). We then compare these measures with the standard deviation calculated over the continuous sample.

The continuous latent variable model has two principal advantages. First, by assuming the existence of a continuous variable, we can model individual responses using well-defined continuous distributions. Second, and more importantly, there is already a well-respected measure of dispersion for continuous variables: the standard deviation. If we treat the continuous variable as a more truthful description of respondents' positions, then the continuous standard deviation is an accurate summary of concentration (or dispersion) in the sample.[1] In the following analysis, ordinal measures that agree more closely with the continuous standard deviation are considered to be more valid.

Because the width of one category – say 0 to 20 for category 1 – is not always substantively equivalent to the width of another category, we also conduct a logarithmic transformation on the simulated points,[2] which treats lower points as substantively further

---

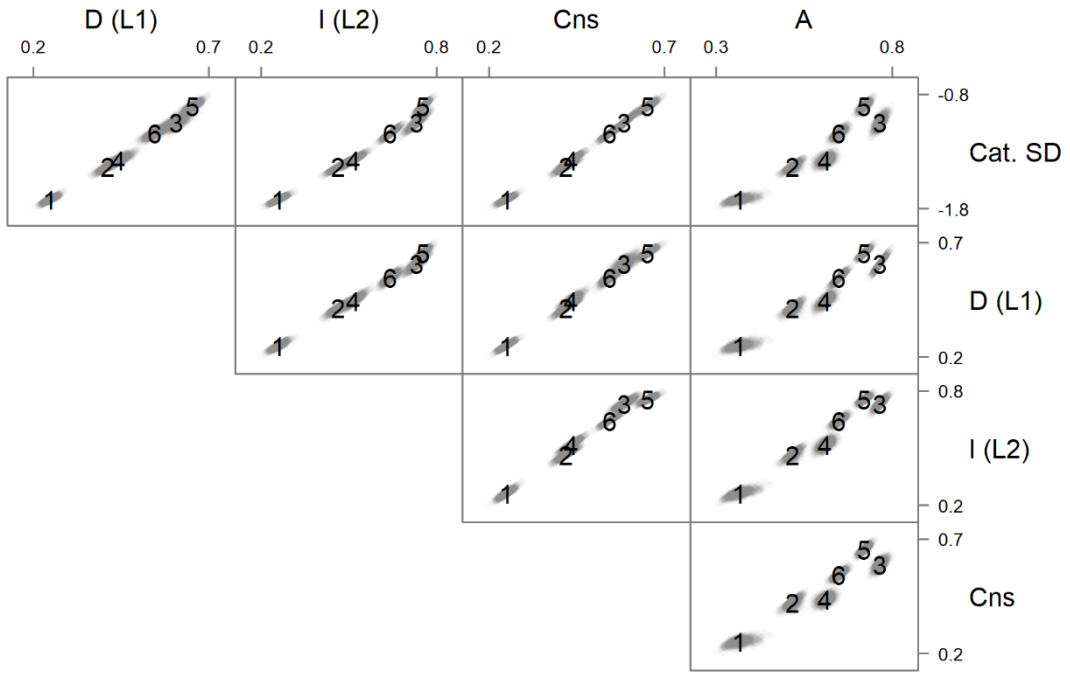
[1] We recognize that there are other measures of statistical dispersion, but the following focus on the most widely used measure.

[2] An alternative approach would be to randomly cut the continuous distribution, but we do not expect to gain additional insight from this computationally intensive approach.

apart (see also Blair and Lacy 2000). We then apply the different measures and the categorical standard deviation to the transformed continuous points.

Initially, we draw on ideal-typical cases to offer examples of the extent to which different indices agree in their measurement of concentration. We consider the six ideal-typical distributions outlined in **Figure 3** above. **Figure 4** graphically shows the different values obtained according to the different measures and how they relate to one another. We can observe high correlations in these ideal-typical cases, but even though the four measures of concentration are standardized to the same range [0, 1], the scores vary widely by measure. For the wide unimodal distribution (Type 4), the indices take values of D=0.571, l=0.697, and A=0.690. They vary even more for the tight unimodal distribution (Type 3), for which Cns=0.398, l=0.479, and A=0.683. Van der Eijk designed A so that it equals 0.5 for the uniform distribution (Type 2), while the other indices vary between D=0.429 and l=0.488.

Figure 4. Comparison of measures in simulated cases



*Notes: The numbers in each panel indicate the ideal type, as outlined in **Figure 3**: 1 bimodal, 2 uniform, 3 highly skewed, 4 skewed, 5 normal with peak, 6 normal but flatter. The scale differs for better readability. The gray areas behind the numbers show the distributions from 1,000 bootstraps as an indication of uncertainty. The four measures D, l, Cns, and Van der Eijk are standardized to [0, 1]. The top-left panel indicates the concentration level according to measure D for the six different ideal-typical distributions.*

Figure 4 demonstrates that it is difficult to determine the concentration of a single distribution based on one of the approaches alone. To a substantial extent, this reflects the fact that, – except for *A* at 0.5 – the different measures provide little intuitive meaning beyond the endpoints 0 and 1. This problem is further complicated by the fact that some authors, such as Blair and Lacy (2000), prefer square indices. While squaring an index will result in a measure that is more sensitive to change when the measure is greater than 0.5, it does not help in understanding a single distribution. In the bimodal distribution (Type 1), for example, l^2=0.051, which ostensibly suggests that the distribution is highly dispersed. The other indices are all around 0.2, suggesting less dramatic dispersion. It follows that

caution should be exercised when employing an index to describe the concentration of a single distribution in *absolute* terms.

Subsequently, we focus on the *relative* performance of the different measures. In **Figure 4**, we can see that the spacing between the areas is inconsistent. Moreover, while most measures provide the same order (1, 2, 4, 6, 3, 5 starting at the bottom in each panel), *A* yields an order of 6-5-3. Measure *A* also differs notably from the others in that it clearly distinguishes between flat distribution (Type 2) and mild concentration (Type 4). We can imagine that preferences for these properties depend on the research question at hand. For the more concentrated distributions (Types 3, 5, 6), we note different sensitivity: L2 considers Types 3 and 5 more similar, *Cns* identifies more similarity between Types 5 and 6, while the standard deviation and L1 space them equally. We expect that in real-world data, it will be difficult to gage the difference or similarity in concentration from the scores alone.

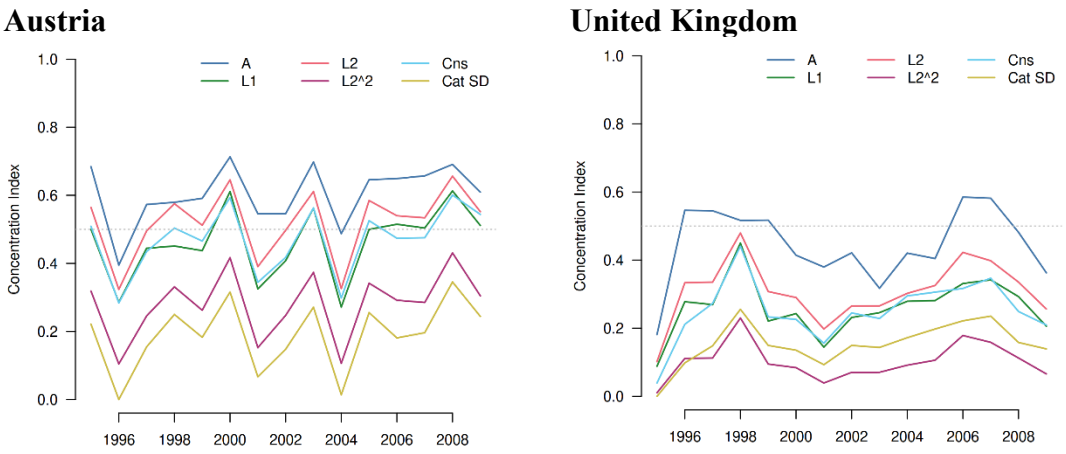
4. Empirical Example: Polarization of political claims

To better understand what the simulations mean in practice, we apply the different measures to data presented by Van der Brug et al. (2015) describing the position of actors toward immigrants in Western European newspaper articles. This introduces data such as those used by other researchers in their own studies and allows us to test whether the results from the ideal-typical cases apply more widely. We calculate the concentration of sentiment by year and country, allowing us to observe trends in concentration between 1995 and 2009. Although data are given for seven European countries, we compare two

in the main text: Austria, in which the indices correlate closely, and the United Kingdom, in which the indices imply different longitudinal trends.

The authors coded claims about immigration in newspapers and assigned a positional score to each claim to identify whether the claim was positive or negative, using 5 categories (Van der Brug et al. 2015). They then calculated Van der Eijk's measurement *A* for claims in a country and year to trace polarization over time. In **Figure 5**, we reproduce two of these figures with different measures (see **Appendix 6** for the other 5 countries, which constitute intermediate cases).

Figure 5. Comparison of Concentration in Austria and the United Kingdom



Notes: Concentration of position of claims in the national newspapers for Austria (left panel) and the United Kingdom (right panel), 1995 to 2009, measured using 5 response categories and different measures of concentration.

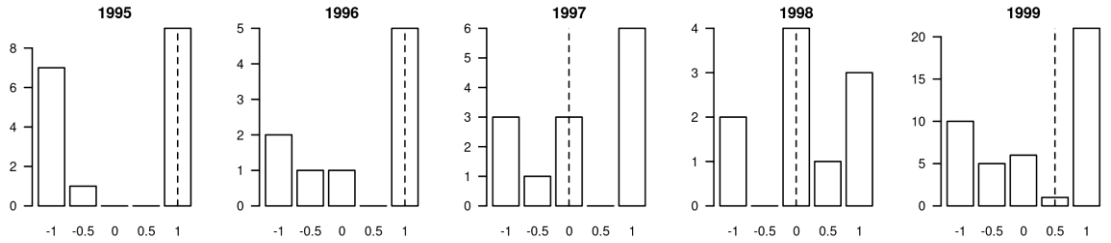
We note two aspects. First, the authors interpreted the values from Van der Eijk's measure *A* using the dotted line at 0.5 as a reference indicating neither concentration nor dispersion – or in their terms: polarization and agreement. This comparison, sanctioned by Van der Eijk (2001), allows them to identify tendencies toward concentration and dispersion in absolute terms. However, the placement of the lines in **Figure 5** makes it clear that such

an interpretation is not possible with the other measures. More generally, looking at the height of the lines in the figure gives us little sense of whether in a particular year media reporting was particularly concentrated or dispersed. Second, in Austria – given on the left of **Figure 5** – the different indices correlate closely, while in the United Kingdom – given on the right of the figure – the indices imply different longitudinal trends. For instance, while Van der Eijk's measure *A* suggests no substantive change in concentration between 1996 and 1999 (blue line at the top), the other indices suggest a peak in concentration in 1998. Inspecting the graph, we can identify a general decrease between 1997 and 2003 using *A*, but a decrease within the notably shorter 1998 to 2001 period. If these values are then related to events such as elections or refugee flows, substantially different conclusions can ensue.

Figure 6 shows the distribution of the positions of claims in the UK between 1995 and 1999 to understand why the different measures of concentration disagree on the changes over time. The situation in 1995 corresponds to a Type 1 distribution with clear polarization (close to case A in **Figure 1**). In 1996 and 1997, there was a concentration of claims with position +1, corresponding to Type 3. In this case, the categorical standard deviation, *A*, and L1 and L2 scores suggest a slight difference in concentration. In contrast, the measure *Cns* identifies a marked difference, with more concentration in 1997. In 1998, there was a concentration at position 0, with secondary peaks of roughly equal size at +1 and -1. In this case, *A* differs from the other measures. Compared with 1997, these two measures identify a slight decrease in concentration, while the others suggest a slight increase. In 1999, the distribution of position was back at Type 3 with a concentration at position +1. Yet, despite the visual similarity to 1996, measures L1 and L2 clearly differ.

We suspect that the manner in which different measures of concentration respond to subtle differences in distributions may not be obvious and readily predictable to social scientists.

Figure 6. Concentration in the UK, 1995 to 1999



Notes: Positions on immigration are measured in 5 categories: -1, -0.5, 0, +0.5, +1. Horizontal dashed lines indicate the median position.

5. Discussion and Conclusion

Ordinal data are common in the social sciences – just think of the Likert items in surveys – and so are questions about concentration and dispersion. It may be suboptimal to use the standard deviation on ordinal data because doing so assumes even spacing between response categories and because the standard deviation is easily influenced by outliers in highly skewed distributions. For this reason, researchers across several fields have proposed measures of concentration and dispersion with ordinal data; however, there has been little work to compare these methods. Here we compare the different approaches to better understand their properties and how they behave in practice. Overall, we found high correlations between the measures, especially when considering ideal-typical cases. However, minor differences in distributions can yield distinct conclusions. For instance, the conclusions for the 1998 UK politicization data differ across measures; and in the simulations, whether we conclude that Type 3 is more concentrated than Type 5 depends on the measure used. Because some of these differences depend on where we are on the

scale between concentration and dispersion, we suspect that some changes are difficult for researchers to anticipate.

Starting with the high correlations, this includes the standard deviation dismissed by the authors of alternative measures for both constructed and real research data. This remains the case when we use the logarithm to vary the spacing between categories – a key argument in favor of alternative measures. In this sense, researchers who gloss over the dangers of the standard deviation when applied to ordinal data may not necessarily draw incorrect conclusions about the concentration of their data: typically, lower standard deviations correspond to higher concentrations. More generally, the different measures tend to agree on which distributions are more concentrated, but they do not agree on the extent of concentration. In other words, if we consider a *change* from a concentrated to a dispersed distribution, the different measures agree that change occurred (because they are all consistent in that they react to shifting 'mass' toward the poles), but they do not agree about the size of the steps. As we have shown, the different measures typically lead to the similar rankings – although measure *A* can differ –, but not the same spacing nor the same scores outside of the endpoints.

Even though (apart from the standard deviation) the measures use the same scale from 0 to 1, we warn researchers against interpreting the scores in absolute terms. Because of the different spacing and because they react differently to minor changes in the distributions, the absolute positions have no intuitive meaning beyond the endpoints (*A* is an exception with an interpretable midpoint at 0.5). Instead, the different measures should be reserved for the comparison of distributions in relative terms: They are reliable in identifying which of the two distributions is more dispersed.

Scores from different measures cannot be compared, nor can we compare scores from 5-point and 7-point scales. In practice, this means that recoding ordinal data (for example if there is some observations in some of the categories) can have a direct impact on the conclusions we draw in terms of concentration and dispersion. If categories are recoded, we recommend comparing concentration and dispersion before and after recoding and discuss differences if they occur.

Depending on the conceptualization or understanding of concentration and dispersion in a particular project, researchers may want to prioritize one of the measures. For example, if we want to measure the reaction to small changes away from the maximum dispersion, L2 is preferable, but this feature does not apply symmetrically for small changes away from the maximum concentration. If a change from a flat distribution to a mild concentration is important, measure *A* emphasizes these more than the others. Measure *A* also has a midpoint that is interpretable in absolute terms; however, the definition of the midpoint may not align with the way researchers understand concentration and dispersion in all applications. Finally, the distinction between exponential and unimodal distributions is a central part of the understanding of concentration. A project should avoid L2, whereas *Cns* is less suited for distinguishing between unimodal distributions with different variance.

In many cases where prioritization is not apparent, as well as tests of robustness, we recommend comparing multiple measures and discussing cases where they differ from one another in qualitative terms. This is necessary because they react differently to changes in distributions; therefore, the measures disagree whether differences between distributions are substantially important or statistically significant for researchers concerned with

this measure. We see no obvious reason to exclude the standard deviation from the suite of indices used to analyze differences in ordinal concentration.

Based on our analysis, we provide the following recommendations:

- All measures of concentration and dispersion considered are suitable for describing changes in distributions, including the standard deviation with which most readers are probably familiar.
- Concentration and dispersion scores have little intuitive meaning beyond the ideal-typical endpoints (and the midpoint in the case of measure *A*). It follows that interpretation in absolute terms should be avoided and that the measures should instead be used to compare two sets of responses on the same scale. In most cases, the choice of measure does not affect broad conclusions on *changes* in concentration and dispersion; therefore, we see no need to crown a single winner.
- Despite the high correlations between scores, we note that differences occur in specific cases; therefore, in the absence of clear prioritization, we recommend using multiple measures of concentration at the same time – coupled with visual inspection of distributions where the measures disagree (i.e., use graphics of the distributions) for a better qualitative understanding of concentration and dispersion.
- The use of multiple measures is also justified by the observation that different measures do not react equally to the same changes in distributions.

- Scores from different measures should not be directly compared. They are not interchangeable, despite high correlations.
- Scores from a distribution with a different number of categories (e.g., 5-point versus 7-point scale) should not be compared.

When considering different measures, we consider it desirable to have hard boundaries – something where the standard deviation falls short – but we have seen that only the ideal-typical endpoints have consistent values across the measures, in which case we do not really need numerical summaries. For the intermediate situations, however, the different measures make it hard to tell how close we are to the ideal points of concentration and dispersion, despite all being able to identify changes in the distribution. Even in cases where the measures are used to compare distributions, a manual examination of the distributions helps understand the behavior of the indices.

While measure A is unique among those considered by providing a clearly defined midpoint, we note that a uniform distribution considered as the midpoint need not necessarily be a relevant cut-off. Instead, we suggest that the definition of concentration and dispersion may indeed depend on the topic under study. This is another reason to visually inspect distributions, even when concentration and dispersion measurements are used to compare distributions.

To conclude, no single measure of concentration and dispersion can serve all cases, and we must be careful with interpretations. In this sense, it is important to understand the limits of the different measures and how they react to different changes in distributions. While for broad changes the different measures will typically yield the same substantive

results, unless researchers have reasons to prioritize one of the measures for how they react to changes in concentration, we recommend using a range of measures of concentration or dispersion and visual assessments, especially in cases where measures disagree in the extent of difference.

References

Alwin, Duane F., and Paula A. Tufiş. 2016. 'The Changing Dynamics of Class and Culture in American Politics: A Test of the Polarization Hypothesis'. The ANNALS of the American Academy of Political and Social Science 663 (1): 229–69.

Balietti, Stefano, Lise Getoor, Daniel G. Goldstein, and Duncan J. Watts. 2021. 'Reducing Opinion Polarization: Effects of Exposure to Similar People with Differing Political Views'. Proceedings of the National Academy of Sciences 118 (52).

Berg, Mark T., Eric A. Stewart, Endya Stewart, and Ronald L. Simons. 2013. 'A Multilevel Examination of Neighborhood Social Processes and College Enrollment'. Social Problems 60 (4): 513–34.

Berry, K., and P. Mielke. 1992. 'Assessment of Variation in Ordinal Data'. Perceptual and Motor Skills 74(1): 63–66.

Blair, J., and M. Lacy. 2000. 'Statistics of Ordinal Variation'. Sociological Methods & Research 28(3): 251–80.

Carvalho, João, and Mariana Carmo Duarte. 2020. 'The Politicization of Immigration in Portugal between 1995 and 2014: A European Exception?' JCMS: Journal of Common Market Studies.

Davies, Vernon. 1970. 'A Re-Examination of Leik's Measure of Ordinal Consensus'. The Pacific Sociological Review 13(1): 27–28.

Enders, Adam M. 2021. "Issues versus Affect: How Do Elite and Mass Polarization Compare?" The Journal of Politics 83(4):1872–77.

Harding, David J. 2011. 'Rethinking the Cultural Context of Schooling Decisions in Disadvantaged Neighborhoods: From Deviant Subculture to Cultural Heterogeneity'. Sociology of Education 84 (4): 322–39.

Krymkowski, Daniel H., Robert E. Manning, and William A. Valliere. 2009. 'Norm Crystallization: Measurement and Comparative Analysis'. Leisure Sciences 31(5): 403–16.

Kubin, Emily, and Christian von Sikorski. 2021. 'The Role of (Social) Media in Political Polarization: A Systematic Review'. Annals of the International Communication Association 45 (3): 188–206.

Kvålseth, Tarald O. 1988. ‘Measuring Variation for Nominal Data’. Bulletin of the Psychonomic Society 26(5): 433–36.

———. 1995. ‘Coefficients of Variation for Nominal and Ordinal Categorical Data’. Perceptual and Motor Skills 80(3): 843–47.

Leik, R. 1966. ‘A Measure of Ordinal Consensus’. Pacific Sociological Review 9(2): 85–90.

Levendusky, Matthew S. 2009. “The Microfoundations of Mass Polarization.” Political Analysis 17(2):162–76.

Lupu, Noam. 2015. “Party Polarization and Mass Partisanship: A Comparative Perspective.” Political Behavior 37(2):331–56.

Tastle, W., and M. Wierman. 2007. ‘Consensus and Dissention: A Measure of Ordinal Dispersion’. International Journal of Approximate Reasoning 45(3): 531–45.

Van der Brug, Wouter, Gianni D’Amato, Joost Berkhout, and Didier Ruedin, eds. 2015. The Politicisation of Migration. Abingdon: Routledge.

Van der Eijk, C. 2001. ‘Measuring Agreement in Ordered Rating Scales’. Quality and Quantity 35(3): 325–41.

Appendix 1 – Measures of concentration

Here, we describe the measures researchers have proposed to capture ordinal concentration or dispersion more precisely. The only algorithm we consider is Van der Eijk's (2001) agreement *A*. It has been adopted by researchers such as Van der Brug et al. (2015) and Carvalho and Duarte (2020) in their analysis of immigration politicization, or by Krymkowski et al. (2009) in their study of norm crystallization in outdoor recreation. This algorithm breaks the vector of responses into uniform layers (for details, see **Appendix 2**). These layers are defined by their pattern of contiguous non-empty and empty categories, and by the number of observations in non-empty categories. Each layer is assigned an agreement value based on the length of empty categories in the layer. The sum of these agreement values, weighted by the size of the layer, is the resulting level of agreement.

When it comes to formulaic measures, we denote the number of response categories – the length of the response vector – by L. The total number of responses is R; the number of responses in each category i is referred to by r_i, and the fraction of total responses in category i by p_i. The empirical cumulative mass function $F = \{F_1, F_2, \ldots, F_L\}$ that is often invoked by such formulas is defined by $F_i := \sum_{j \leq i} \frac{r_j}{R} = \sum_{j \leq i} p_j \quad (1 \leq i \leq L)$.

Blair and Lacy (2000) define $F^{\text{maxdisp}} = \left(\frac{1}{2}, \frac{1}{2}, \ldots, 1\right)$, the cumulative mass function corresponding to the point of minimal concentration and maximal dispersion in which the responses are split equally between each extreme. Similarly, we can define $F^{\text{maxconc}} = (0,1,1, \ldots, 1)$ as a cumulative mass function corresponding to one of the points of maximal concentration where all responses fall into a single category. Blair and Lacy (2000) observe that Leik's (1966) *D*, Berry and Mielke's (1992) *IOV*, Kvålseth's (1995) *COV*, and

their (2000) *l* and l^2, all belong to a family of measures of the form $\text{Concentration} = \frac{d\left(F_{-L}, F_{-L}^{\text{maxdisp}}\right)}{d\left(F_{-L}^{\text{maxconc}}, F_{-L}^{\text{maxdisp}}\right)}$, where *d* is a distance function defined on the first *L-1* entries of each cumulative mass vector. We have added the subscript $-L$ to F, F^{maxdisp} and F^{maxconc} to make clear that we are only measuring the first *L-1* entries in these vectors. Dividing by $d\left(F_{-L}^{\text{maxconc}}, F_{-L}^{\text{maxdisp}}\right)$ ensures that the resulting index is standardized between 0 and 1. Zero corresponds to maximum dispersion, and 1 corresponds to maximum concentration. The complementary measure *dispersion = 1 – concentration* follows from this definition. Note that, while there are *L* possible maximally concentrated cumulative mass functions F^{maxconc}– corresponding to the *L* possible positions in which all mass can be placed – for each of them the denominator $d\left(F_{-L}^{\text{maxconc}}, F_{-L}^{\text{maxdisp}}\right)$, will be the same under any of the distance functions *d* discussed below.

The precise nature of the distance function, d, varies according to the measure used. Leik's measure uses the L1 distance, commonly known as the 'city-block' distance. This means that Leik's (1966) D can be written as: $D = \frac{L^1\left(F_{-L}, F_{-L}^{\text{maxdisp}}\right)}{L^1\left(F_{-L}^{\text{maxconc}}, F_{-L}^{\text{maxdisp}}\right)} = \frac{\sum_{i=1}^{L-1}|F_i - 1/2|}{(L-1)/2}$. Blair and Lacy (2000) define a measure *l* of concentration which uses the L^2 distance. Thus, *l* is written as: $l = \frac{L^2\left(F_{-L}, F_{-L}^{\text{maxdisp}}\right)}{L^2\left(F_{-L}^{\text{maxconc}}, F_{-L}^{\text{maxdisp}}\right)} = \left[\frac{\sum_{i=1}^{L-1}\left(F_i - \frac{1}{2}\right)^2}{\frac{(L-1)}{4}}\right]^{\frac{1}{2}}$. Berry and Mielke's (1992) *IOV* measure of dispersion is equal to $1–l^2$, while Kvålseth's (1995) *COV* measure of dispersion is equal to *1–l*.

Expanding on the Shannon entropy formula, Tastle and Wierman (2007) define a measure of consensus *Cns* that does not explicitly rely on the cumulative-distribution approach.

Instead, their measure incorporates the sample mean $\overline{X}$ of the responses X_i: $Cns(X_1, X_2, \ldots, X_R) = 1 + \sum_{i=1}^{R} p_i \, log_2 \left(1 - \frac{|X_i - \overline{X}|}{d_X}\right)$. Here, $d_X \coloneqq \max(X) - \min(X)$ is the width of the categories (Tastle and Wierman 2007). A question with response categories 1, 2, and 3 has a length of 3–1=2.

Although it does not invoke the cumulative distribution function, Tastle and Wierman's (2007) *Cns* measure relies on $\overline{X}$, which treats indices of categories as absolute positions with equal distance between them. In addition, Davies (1970) remarked that, because it combines linearly the cumulative distribution function *F*, Leik's measure *D* assumes that the categories are equally spaced. It is possible that such "supra-ordinal assumptions" (Davies 1970, 27), may affect the validity of other measures using *F*. For these reasons, we compare all measures to the standard deviation calculated over the category indices. Measures that are indistinguishable from the categorical standard deviation may indeed make supra-ordinal assumptions; moreover, given the choice between a supposedly ordinal measure and the categorical standard deviation, it makes sense to choose the simpler one if both formulas give comparable results.

Blair and Lacy (2000) offer analytical measures of variance of the *l* index for both small and large samples, permitting the construction of analytical confidence intervals. Despite this, hypothesis testing remains difficult to conduct. Blair and Lacy note that the null hypothesis that *l*=*r*, for example, would describe many points with L^2 distance *r* from F_{-L}^{maxdisp}, a problem that increases in difficulty as the number of response categories increases (ibid.). Attempting a hypothesis test with Leik's *D* results in the same difficulty. Tastle and Wierman (2007) do not obtain an analytical measure of variance for *Cns*, nor

does Van der Eijk (2001) for the measure *A*. To overcome the insufficient development of analytical measurements of variance of these indices, we use the bootstrap method to see whether the differences in concentration between different values of dispersion or concentration are statistically reliable. In each analysis, we compare Blair and Lacy's *l*, Leik's *D*, Tastle and Wierman's *Cns*, Van der Eijk's *A*, and the categorical standard deviation. Although we can study the correlations between these measures, there is no intuitive way to determine which measures are more accurate. To resolve this challenge in the simulations, we suppose that ordinal responses correspond to an underlying, continuous latent variable. Variants of this assumption are central in many ordinal models, such as the ordinal logistic regression model.

Appendix 2 – Calculating *A*

The following describes how the agreement score is calculated by Van der Eijk (2001). The frequency vector represents a 7-point ordinal variable: (30, 40, 210, 130, 530, 50, 10). This is divided into layers where each response category has the same number of observations. The first layer has the size of the smallest category (n=10), and no empty categories (pattern 1111111). The number of non-empty categories in the pattern is S=7. The weight of the layer is $\frac{\text{L=7 } categories * \text{n=10 } observations}{R\text{=1000 } responses \text{ } in \text{ } the \text{ } vector} = 0.07$. The agreement of each layer is calculated as $A_{layer} = 1 - \frac{S-1}{L-1}$, where S is the number of non-empty categories in a layer, and L is the total number of response categories (L=7 in this example). For layer 2, the number of non-empty categories in the pattern is S=6. The agreement of the layer is thus 0.0167. The agreement score *A* for the entire vector is the weighted average of the layer agreements. We refer the reader to the original paper for a more thorough explanation of the algorithm.

Table A1. Van der Eijk's algorithm divides the vector into layers, and identifies non-empty categories in each layer as the basis of a weighted agreement score

	1	2	3	4	5	6	7	Pattern	Layer agreement	Weight
Layer 1	10	10	10	10	10	10	10	1111111	0	0.07
Layer 2	20	20	20	20	20	20	0	1111110	0.167	0.12
Layer 3	0	10	10	10	10	10	0	0111110	0.333	0.05
Layer 4	0	0	10	10	80	10	0	0011110	0.5	0.04
Layer 5	0	0	80	80	80	0	0	0011100	0.667	0.24
Layer 6	0	0	80	0	80	0	0	0010100	0.467	0.16
Layer 7	0	0	0	0	320	0	0	0000100	1	0.32
Total	30	40	210	130	530	50	10			

Appendix 3 – Additional distributions

Distributions from Tastle and Wierman that are scaled versions of each other or differ only marginally (distribution O) are not included in **Figure 1**. **Table A2** shows that these (by definition) only vary in the measures derived from the standard deviation.

Table A2. Concentration scores for the rescaled distributions

	A	L1	L2	L2^2	Cns	SD	SD (log)
K	0.000	0.000	0.000	0.000	0.000	2.108	0.848
L 	0.000	0.000	0.000	0.000	0.000	2.010	0.809
M 	0.000	0.000	0.000	0.000	0.000	2.001	0.805
N 	0.000	0.000	0.000	0.000	0.000	2.000	0.805
O 	0.020	0.020	0.020	0.000	0.000	2.010	0.809
U	0.792	0.750	0.791	0.625	0.686	1.000	0.347
V 	0.792	0.750	0.791	0.625	0.686	0.877	0.304
W 	0.792	0.750	0.791	0.625	0.686	0.871	0.302
X 	0.792	0.750	0.791	0.625	0.686	0.870	0.301
Y 	0.792	0.750	0.791	0.625	0.686	0.866	0.300

*Notes: Rescaled distributions that are not included in Figure 3. The highest bar in distribution K is 5 (the histogram is scaled differently from **Table 1**), in distribution L 50, in distribution M 500, in distribution N 5000. Distribution O resembles distribution L, but one bar is 2 larger than the other. For distribution U, the non-zero bars are 1 and 3, for distribution V 10 and 30, for distribution W 20 and 60, for distribution X 30 and 90, and for distribution Y 300 and 900.*

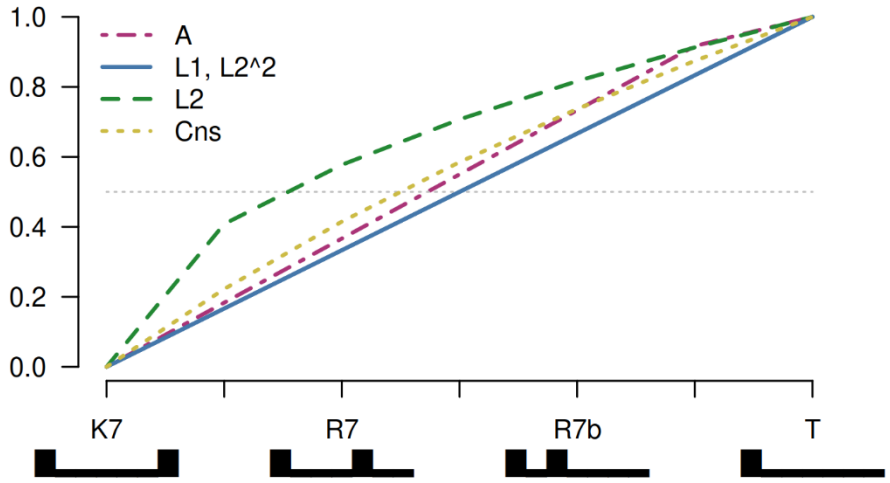
Appendix 4 – Seven categories

Change from maximum dispersion to maximum concentration for 5 categories, and for 7 categories. The cases correspond to those presented in **Figure 1**, with the number 7 highlighting the use of 7 categories. Where the use of 7 categories required a distinction, these are indicated with letters ‘a’ and ‘b’.

Table A3. Concentration scores for distributions based on 7 categories

Case	A	L1	L2	L2^2	Cns
K7	0.000	0.000	0.000	0.000	0.000
Q7	0.183	0.167	0.408	0.167	0.222
R7	0.367	0.333	0.577	0.333	0.415
R7a	0.550	0.500	0.707	0.500	0.585
R7b	0.733	0.667	0.816	0.667	0.737
S7	0.917	0.833	0.913	0.833	0.874
T7	1.000	1.000	1.000	1.000	1.000
A7	0.000	0.000	0.000	0.000	0.000
B7	0.383	0.167	0.289	0.083	0.208
B7b	0.473	0.333	0.430	0.185	0.384

Figure A1. Changes in concentration for 7 categories



Notes: compare **Figure 2** for 5 categories.

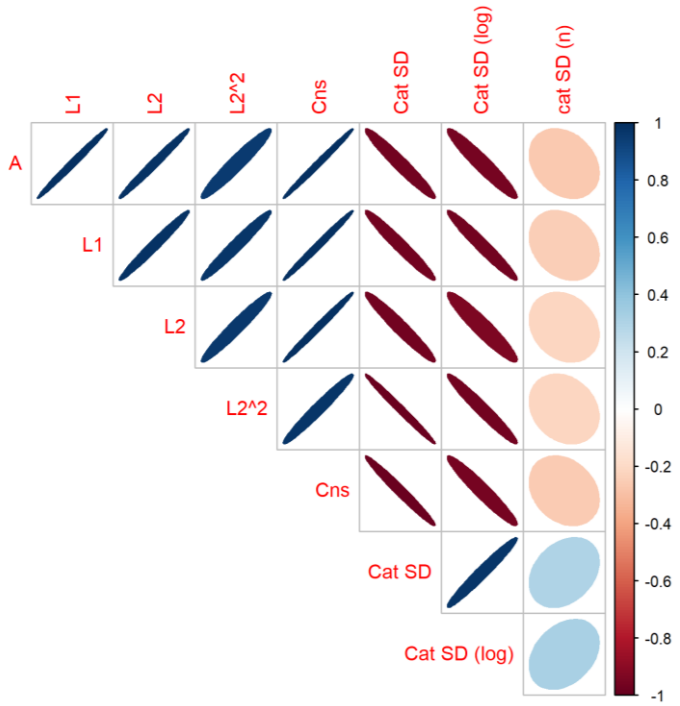
Appendix 5 – Correlations

Table A2. Correlation between Measures of Concentration

	A	L1	L2	L2^2	Cns	SD	SD (log)
A	1.00	0.99	0.99	0.96	0.99	-0.96	-0.95
L1	0.99	1.00	0.98	0.97	0.99	-0.97	-0.97
L2	0.99	0.98	1.00	0.96	0.99	-0.96	-0.94
L2^2	0.96	0.97	0.96	1.00	0.97	-0.99	-0.97
Cns	0.99	0.99	0.99	0.97	1.00	-0.98	-0.96
SD	-0.96	-0.97	-0.96	-0.99	-0.98	1.00	0.97
SD (log)	-0.95	-0.97	-0.94	-0.97	-0.96	0.97	1.00
SD/n	-0.26	-0.25	-0.21	-0.22	-0.26	0.31	0.33

Notes: based on cases A to Y, Pearson correlation coefficients.

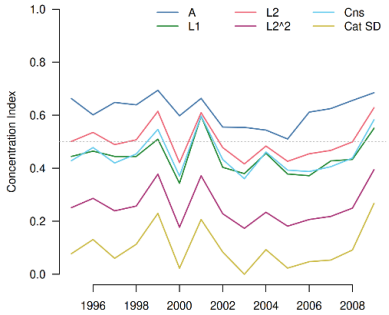
Figure A2. Correlation between Measures



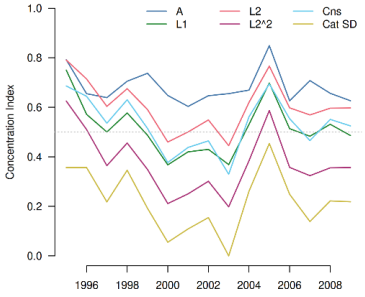
Appendix 6 – Country distributions

Figure A3. Polarization for Belgium, Ireland, Netherlands, Spain, Switzerland, 1995 to 2009

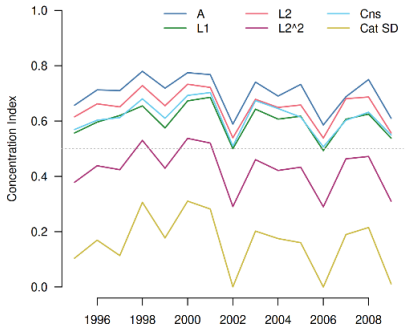
Belgium



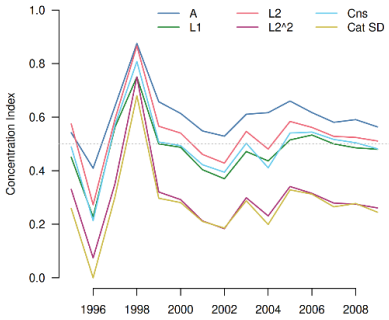
Ireland



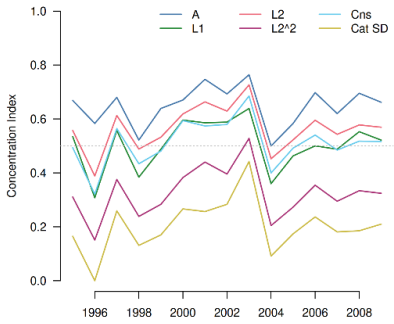
Netherlands



Spain



Switzerland



Notes: refer to **Figure 5** for a description